

Chapter 11 | Gases and the Gas Laws

Chemistry I · Unit 4: States of Matter · Grade 11

11.1 The Behavior of Gases

A gas has no fixed shape and no fixed volume. Its particles move freely and independently, colliding with each other and with the walls of the container. Because those particles are so far apart relative to their size, a gas can be compressed into a much smaller space than a liquid or a solid. This is why the volume of a gas is not a property of the gas alone. It depends on the conditions the gas is held under.

Four measurable quantities describe the state of any gas sample: pressure, volume, temperature, and the amount of gas in moles. If you know three of them, the fourth is determined. The relationship between them is the subject of this chapter.

11.2 Pressure and Volume

Robert Boyle observed in 1662 that when the temperature of a fixed amount of gas is held constant, increasing the pressure decreases the volume. The two quantities are inversely proportional. Doubling the pressure halves the volume. Reducing the pressure to one third triples the volume.

$$P_1 V_1 = P_2 V_2$$

The physical picture is a piston in a sealed cylinder. Push the piston down and the same number of particles are confined to a smaller space, so they strike the walls more often and the pressure rises. Release the piston and the gas expands until the pressure inside matches the pressure outside.

11.3 Temperature and Volume

Jacques Charles found that when pressure is held constant, the volume of a gas is directly proportional to its absolute temperature. Heating a gas makes its particles move faster. Faster particles strike the walls harder and more often, and the gas pushes outward until it occupies more space.

$$V_1 / T_1 = V_2 / T_2$$

Temperature in every gas law must be expressed in kelvin. To convert, add 273.15 to a Celsius reading. Using Celsius directly will produce nonsense, including negative volumes.

11.4 The Ideal Gas Law

The separate observations of Boyle, Charles, and Avogadro combine into one equation that relates all four quantities at once. This is the ideal gas law, and it is the central result of the chapter.

$$PV = nRT$$

Here P is pressure, V is volume, n is the amount of gas in moles, T is the absolute temperature in kelvin, and R is the universal gas constant. When pressure is measured in kilopascals and volume in liters, R has the value 8.314 L·kPa per mol·K.

Solving for volume gives the form used most often in laboratory work, because pressure and temperature are the two conditions an experimenter can adjust directly:

$$V = nRT / P$$

Read this equation as a statement about competition. Temperature pushes the volume up. Pressure pushes it down. The volume you actually observe is the result of the two effects working against each other. Raising the temperature and raising the pressure by matching proportions leaves the volume unchanged.

11.5 Worked Example

A sealed cylinder contains 1.00 mol of nitrogen gas. The pressure is set to 100 kPa and the temperature to 300 K. Find the volume.

$$V = nRT / P = (1.00)(8.314)(300) / 100 = 24.9 \text{ L}$$

Now suppose the experimenter needs the gas to occupy exactly 18.0 L. Several combinations of pressure and temperature will produce that volume. Lowering the temperature to 216 K at the same pressure would do it. So would raising the pressure to 138 kPa at the original temperature. There is no single correct answer, only a set of conditions that satisfy the equation.

11.6 Limits of the Ideal Model

The ideal gas law assumes gas particles have no volume of their own and exert no attractive forces on one another. Real gases follow it closely at ordinary pressures and temperatures well above their boiling points. The model breaks down at very high pressure, where particle volume stops being negligible, and at very low temperature, where attractive forces begin to matter.

Practice

1. A gas occupies 12.0 L at 250 kPa. What volume will it occupy at 150 kPa if temperature is unchanged?
2. A 2.00 mol sample is held at 400 K and 200 kPa. Calculate the volume.
3. An engineer needs 1.00 mol of gas to fill exactly 20.0 L. Find two different pairs of pressure and temperature that achieve this, and explain why more than one answer exists.
4. Explain, in terms of particle motion, why raising the temperature of a gas in a rigid sealed container raises the pressure rather than the volume.